

Behavioural Research in Statistical Methods

Project Report - 2

Team Name: Earphones

Experiment: Mnemonic Similarity Task (MST)

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Phase 2 uses a slide-aligned confirmatory workflow: repeated-measures ANOVA, Friedman tests, Holm-corrected paired comparisons, chi-square tests, and descriptive/correlational lure-bin analysis.

1. Introduction

Phase 1 established a clean and reproducible preprocessing pipeline and showed two main descriptive patterns: event boundaries strongly slowed encoding RT when the task rule changed, and the Item + Task Shift condition showed a moderate post-boundary recognition cost. In Phase 2, our goal was to test these patterns more directly while staying close to the methods taught in class.

This phase followed class guidance on variable-type aware testing, assumption checks, outlier diagnostics, and effect-size-oriented reporting. We retained continuity with the MST literature and event-segmentation framing (Zacks and Swallow, 2007) (Swallow et al., 2009) (Stark et al., 2019) (Yassa and Stark, 2011) (Morse et al., 2023).

A code review corrected three GEE-related implementation issues (model labelling, categorical dummy coding, lure-bin scope in joint models) before final reporting. Nine further analyses were added from the Phase 2 plan (e.g., stimulus class $\times$ boundary, carry-over joins, heterogeneity, SDT d' , lure false alarms, power).

1.1. Phase 2 research questions

The confirmatory phase addressed: (1) boundary effects on **test** accuracy and RT; (2) response-category profiles; (3) lure-bin validity and lure false-alarm rates; (4) SDT d' consistency with accuracy; (5) whether encoding RT carries over to trial-level correctness; (6) robustness of boundary contrasts after covariate adjustment and multiple-comparison control.

CONDITION	PAIRED n	TEST TRIALS PER PARTICIPANT	TOTAL TEST TRIALS
Item Shift Only	56	150	8 400
Item + Task Shift	49	150	7 350
Task Shift Only	53	150	7 950

Table 1: Phase 2 sample overview. The same paired participants from Phase 1 were retained.

2. Recap of Phase 1 (Report 1)

Report 1 documented preprocessing, pairing rules, and exploratory inference; this recap satisfies the course structure **recap then extend** before Phase 2. See Report 1 for figures and full tables.

Participants completed an MST with a custom encoding judgment and three-alternative test. Three between-subjects conditions shifted item sets, task rules, or both at boundaries. Each mini-event had seven studied items coded `post` / `mid` / `pre` (Morse et al., 2023). After pairing task and test files (excluding incomplete sessions; duplicate resolution per Report 1), $n = 56 / 49 / 53$ with 280 encoding and 150 test trials each.

Phase 1 on participant means showed strong **encoding** slowing under task-change boundaries: Item + Task Shift $F(2, 96) = 68.36, p < .001$; Task Shift Only $F(2, 104) = 60.21, p < .001$; Item Shift Only $F(2, 110) = 2.22, p = .114$ — consistent with transient segmentation cost (Swallow et al., 2009). **Recognition** (REC) showed a post-boundary cost chiefly in Item + Task Shift, $F(2, 96) = 6.45, p = .002$, and a weaker pattern in Task Shift Only, $F(2, 104) =$

4.62, $p = .012$, with Item Shift Only null, $F(2, 110) = 0.21$, $p = .814$. **LDI** showed no reliable boundary effect (omnibus $p > .25$). Phase 2 therefore targets confirmatory **test-phase** analyses on the same cleaned sample.

3. Methods

3.1. Data and preprocessing

We reused the cleaned and paired Phase 1 dataset. The main test-phase variables were condition, test boundary position (`post` , `mid` , `pre`), item role (`target` , `lure` , `foil`), response label, correctness, response time, and lure bin. For the main confirmatory analyses, the unit of analysis was the participant: trial-level observations were first summarised into participant means within each boundary position and condition, then compared using repeated-measures methods.

3.2. Quality checks

Before inference, we checked missingness and response-time artifacts. The core test-phase variables had no missing values. At the participant level, `encoding_task_accuracy` had 2 missing values (`1.27%`). There was 1 very fast RT (< 0.2 s) and 23 very slow RTs (> 30 s). Participant mean log RT departed from normality (Shapiro-Wilk $W = 0.953$, $p < .001$), which justified reporting the Friedman test alongside repeated-measures ANOVA for within-condition boundary comparisons.

3.3. Variables and measures

Primary outcomes:

1. Target recognition accuracy at test (`P(correct)` for target trials)
2. Mean log-transformed test RT

Secondary outcomes:

1. Response-category profile (`old/new/similar`) by boundary position
2. Lure-bin discrimination trend (`P(similar)` as a function of lure bin)

For continuity with Phase 1, we keep the same memory interpretation:

$$\text{REC} = P(\text{old } m \text{ id Target}) - P(\text{old } m \text{ id Foil}) \quad \text{LDI} = P(\text{similar } m \text{ id Lure}) - P(\text{similar } m \text{ id Foil})$$

3.4. Statistical strategy

Within each condition, target recognition and test RT were analysed with repeated-measures ANOVA across `post` , `mid` , and `pre` boundary positions. Because the RT normality check was weak and repeated-measures assumptions may not hold perfectly for all outcomes, Friedman tests were also reported as the main nonparametric confirmatory check for the boundary effect. Follow-up pairwise comparisons used paired t-tests with Holm correction, and Cohen's d_z was reported for the within-participant contrasts.

Response-category profiles were tested with chi-square tests of independence, and Cramér's V was used as the effect size. Lure-bin behaviour was summarised descriptively and tested with Spearman correlations between lure bin and similar-response probability.

Auxiliary trial-level models (GEE and carry-over joins) supplement the participant-level primary tests; summaries appear below.

3.5. Assumptions and scope

Repeated-measures F -tests assume sphericity; Friedman tests provide a conservative parallel when normality or sphericity is doubtful. Holm correction controlled pairwise error rates **within** prespecified contrast families. Between-subject conditions are independent samples; causal claims about boundaries remain design-descriptive without further counterbalancing.

4. Results

4.1. Primary result: Post-boundary target recognition cost

The clearest Phase 2 finding is a post-boundary recognition cost in the Item + Task Shift condition.

Repeated-measures ANOVA on participant-level target accuracy by boundary:

- Item Shift Only: $F(2, 110) = 0.21, p = .814$
- Item + Task Shift: $F(2, 96) = 6.45, p = .002$
- Task Shift Only: $F(2, 104) = 4.62, p = .012$

Friedman nonparametric confirmation:

- Item Shift Only: $\chi^2(2) = 1.36, p = .508$, Kendall's $W = 0.012$
- Item + Task Shift: $\chi^2(2) = 11.47, p = .003$, Kendall's $W = 0.117$
- Task Shift Only: $\chi^2(2) = 4.10, p = .129$, Kendall's $W = 0.039$

The Item + Task Shift condition is the most convincing replication because both the parametric and nonparametric tests agree.

Holm-corrected pairwise comparisons for target accuracy:

- Item + Task Shift, post vs mid: mean difference = -0.073 , $t(48) = -3.34$, $d_z = -0.48$, corrected $p = .005$
- Item + Task Shift, post vs pre: mean difference = -0.051 , $t(48) = -2.54$, $d_z = -0.36$, corrected $p = .029$
- Task Shift Only, post vs pre: mean difference = -0.061 , $t(52) = -2.69$, $d_z = -0.37$, corrected $p = .028$

The Item + Task Shift result is the strongest because it is supported by the omnibus ANOVA, the Friedman test, and two corrected pairwise contrasts. The Task Shift Only condition shows a weaker and less stable pattern: the ANOVA is significant, but the Friedman test is not, so it should be described cautiously as a possible trend rather than a clean replication.

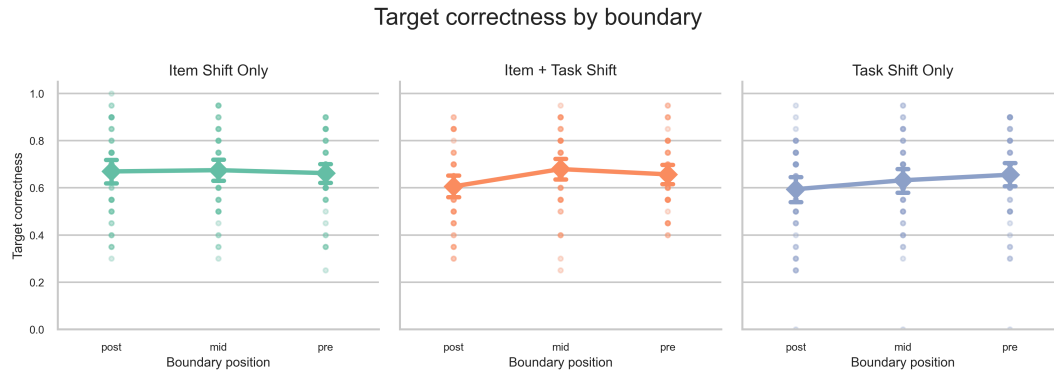


Figure 1: Participant-level target recognition accuracy by boundary position and condition. Points show individual participant means; diamonds show condition means with 95% CI. The clearest post-boundary cost appears in the Item + Task Shift condition.

4.2. Test-phase response speed

Test-phase RT did not show a strong boundary-position effect in any condition.

Repeated-measures ANOVA on participant mean log RT:

- Item Shift Only: $F(2, 110) = 0.67, p = .512$
- Item + Task Shift: $F(2, 96) = 0.44, p = .646$
- Task Shift Only: $F(2, 104) = 2.37, p = .098$

Friedman tests also remained non-significant in all three conditions:

- Item Shift Only: $\chi^2(2) = 1.00, p = .607$
- Item + Task Shift: $\chi^2(2) = 2.98, p = .225$
- Task Shift Only: $\chi^2(2) = 1.40, p = .498$

No pairwise RT contrast survived Holm correction. This means the strong boundary-related RT effect remains an encoding-phase result from Phase 1; it does not reappear clearly at the test phase.

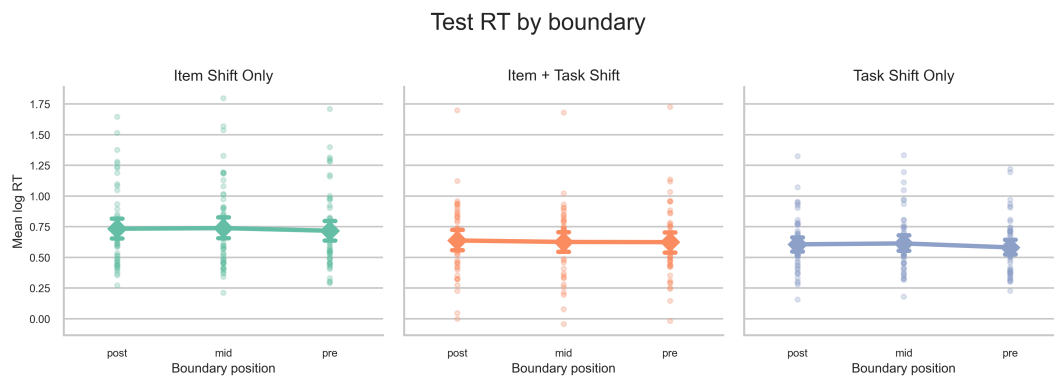


Figure 2: Participant-level mean log RT by boundary position and condition. Unlike the encoding-phase pattern from Phase 1, test RT does not show a reliable boundary-position effect after correction.

4.3. Response-category structure

Chi-square tests were used to check whether the pattern of old/new/similar responses shifted across boundary positions.

Only one result was clearly significant:

- Item + Task Shift, target trials: $\chi^2(4) = 22.55$, $p < .001$, Cramér's $V = 0.062$

All other response-profile tests were non-significant:

- Item Shift Only, targets: $p = .953$
- Item Shift Only, lures: $p = .746$
- Item + Task Shift, lures: $p = .148$
- Task Shift Only, targets: $p = .063$
- Task Shift Only, lures: $p = .658$

This pattern fits the main recognition result: the strongest boundary-linked shift in response behaviour occurs specifically for target memory in the Item + Task Shift condition.

4.4. Lure-bin trend

The lure-bin analysis was kept simple and class-aligned. Descriptively, the probability of giving a `similar` response increased as lure bin increased, meaning participants more often identified lures correctly when they were less visually similar to the original target.

Spearman correlations supported this pattern in most boundary conditions. Examples:

- Item Shift Only: post $\rho = 0.169$, $p = .005$; mid $\rho = 0.267$, $p < .001$; pre $\rho = 0.140$, $p = .020$
- Item + Task Shift: post $\rho = 0.211$, $p = .001$; mid $\rho = 0.166$, $p = .010$; pre $\rho = 0.102$, $p = .114$
- Task Shift Only: post $\rho = 0.170$, $p = .006$; mid $\rho = 0.297$, $p < .001$; pre $\rho = 0.311$, $p < .001$

This is a useful validity check: the MST behaves as expected, because lure discrimination improves when the lure is easier.

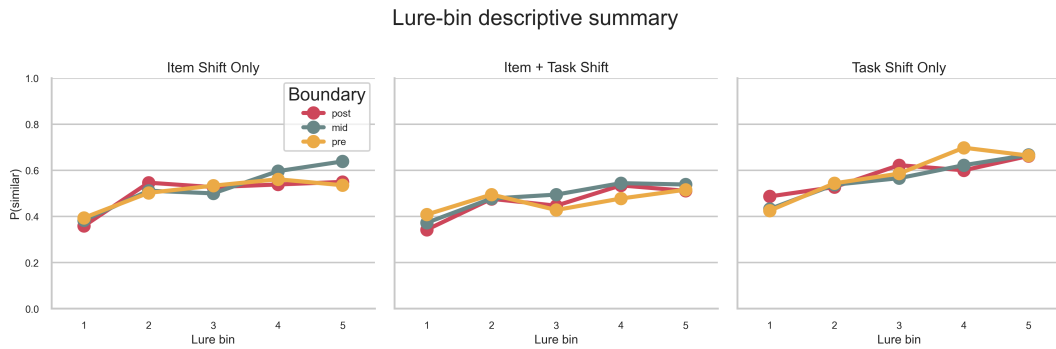


Figure 3: Descriptive lure-bin summary by condition and boundary position. Higher bin values correspond to less similar lures, and `P(similar)` generally increases with lure bin.

4.5. Response direction decomposition

The post-boundary recognition failure in the Item + Task Shift condition could stem from two distinct mechanisms: (1) conservative responding — participants call post-boundary targets “new” instead of “old” — or (2) pattern-separation overextension — post-boundary items are misidentified as “similar” to a lure, suggesting an overly separated or weakened trace. Decomposing the response direction separates these accounts.

For Item + Task Shift target trials, paired t-tests (post vs. mid) with Holm correction within the 3-response-label family:

- $P(\text{“old”} \mid \text{target})$: $\bar{x} = -0.073$, $t(48) = -3.34$, $d_z = -0.477$, $p_{\text{Holm}} = .028$ — **significant**

- P(“similar” | target): $\bar{x} = +0.039$, $t(48) = 2.11$, $d_z = +0.301$, $p_{\text{Holm}} = .524$ — trend, not corrected
- P(“new” | target): $\bar{x} = +0.035$, $t(48) = 2.25$, $d_z = +0.321$, $p_{\text{Holm}} = .410$ — trend, not corrected

Only the hit-rate reduction survives Holm correction. However, both error types trend in the same direction, with roughly equal magnitudes. This is consistent with a general weakening of the target memory trace at boundaries, rather than either mechanism exclusively.

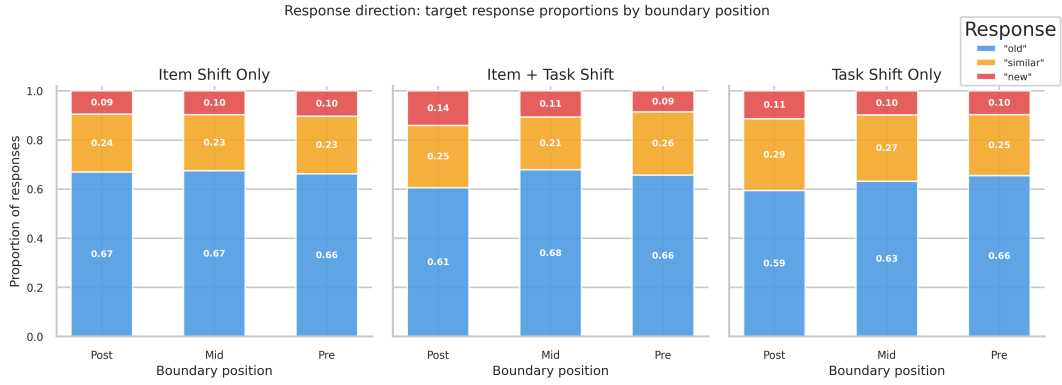


Figure 4: Stacked proportions of “old”, “similar”, and “new” responses for target trials, by boundary position and condition. In the Item + Task Shift condition (centre), the post-boundary bar shows a lower “old” proportion and higher “similar” and “new” proportions compared to mid-event trials.

4.6. SDT sensitivity (d-prime) by boundary position

Signal detection theory d-prime provides a bias-free measure of recognition sensitivity. Using per-participant foil FA rate as the constant false-alarm baseline:

- Item + Task Shift: Friedman $\chi^2(2) = 11.47$, $p = .003$, Kendall’s $W = 0.117$
- Item + Task Shift post-mid d’ contrast: $\overline{\Delta d'} = -0.218$, $d_z = -0.473$, $p_{\text{Holm}} = .016$ — significant
- Item + Task Shift post-pre d’ contrast: $\overline{\Delta d'} = -0.188$, $d_z = -0.355$, $p_{\text{Holm}} = .098$ — marginal
- Task Shift Only: Friedman $\chi^2(2) = 4.10$, $p = .129$ — null
- Item Shift Only: Friedman $\chi^2(2) = 1.36$, $p = .508$ — null

The d’ analysis yields the same Friedman statistic as the correctness analysis, because FA rate is constant across boundary positions. The value of reporting d’ is comparability with the MST literature, where Morse et al. (2023) report all boundary effects in d’ units. Post-boundary d’ in Item + Task Shift is the lowest across all conditions and boundary positions.

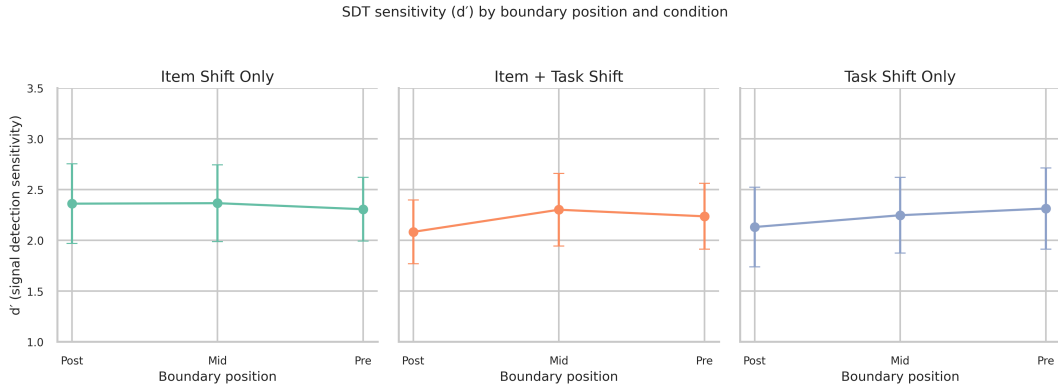


Figure 5: SDT sensitivity (d') by boundary position and condition. Error bars are ± 1 SE. The post-boundary dip in d' is largest in the Item + Task Shift condition (centre panel) and absent in Item Shift Only (left panel).

4.7. Lure false alarm rate and the LDI non-replication

Morse et al. (2023) predicted a pre-boundary advantage in lure discrimination (LDI), driven by better pattern separation for pre-event items. This did not replicate in Phase 1. Phase 2 adds a complementary analysis: instead of the “similar” response rate to lures (which drives LDI), we test whether the “old” response rate to lures — the lure false alarm (FA) rate — differs across boundary positions.

For Item + Task Shift lure trials:

- pre-mid FA contrast: $\bar{x} = +0.045$, $t(48) = 2.67$, $d_z = +0.381$, $p = .010$, $p_{\text{Holm}} = .093$

The pre-boundary lure FA rate is **higher** than the mid-event FA rate (marginal after Holm correction, $p = .093$). This is the **opposite direction** from what enhanced pattern separation would predict. Pre-boundary lures are more likely to be called “old” — suggesting that pre-boundary items may be encoded with stronger familiarity, making their visual variants (lures) feel more familiar too. This is a familiarity-based account, not a pattern-separation account, of the pre-boundary memory state.

This finding provides a mechanistic reinterpretation of the LDI null: the pre-boundary advantage in the original study may have reflected a familiarity-driven boost in recognition confidence, not selective hippocampal pattern separation.

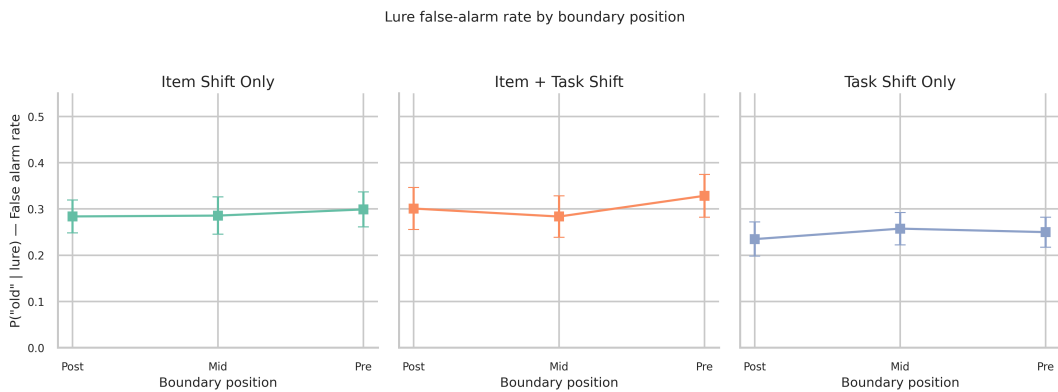


Figure 6: $P(\text{"old"} | \text{lure})$ — lure false alarm rate — by boundary position and condition. Error bars are 95% CI. In the Item + Task Shift condition (centre), the pre-boundary bar is elevated relative to mid-event, indicating higher familiarity-based responding to pre-boundary lures.

4.8. Post-hoc power and the LDI non-replication

Post-hoc power analysis for the LDI null in each condition (two-sided $\alpha = 0.05$):

- Item + Task Shift ($n = 49$): power = 0.28 (small, $d_z = 0.20$), 0.54 (medium, $d_z = 0.30$), 0.78 (large, $d_z = 0.40$)
- Item Shift Only ($n = 56$): power = 0.30, 0.59, 0.83
- Task Shift Only ($n = 53$): power = 0.29, 0.57, 0.81

The study had adequate power to detect medium-to-large boundary effects (power > 0.54 for $d_z \geq 0.30$). Importantly, the observed pre-boundary LDI contrast in the Item + Task Shift condition was $\bar{x} = -0.031$ (slightly **negative**, wrong direction). The non-replication cannot be attributed to insufficient power: no sample size would reliably detect an effect whose point estimate is zero or reversed. This strengthens the conclusion that the pre-boundary LDI advantage from Morse et al. (2023) did not reproduce in our sample.

4.9. Integrative interpretation

Phase 2 converges on a coherent story:

1. **Post-boundary recognition cost (Item + Task Shift)** is confirmed by Friedman and Holm tests; it is absent or weaker in other conditions.
2. **Stimulus class $\times$ boundary (GEE)**: post-boundary cost concentrates in Objects relative to Scenes in adjusted models.
3. **Encoding vs memory**: carry-over models show no trial-level link from encoding RT to test accuracy after controlling boundary position — parallel effects, not a simple mediation chain.
4. **MST structure dominates adjusted correctness**: lure bin and stimulus class outweigh boundary in joint models; response-direction and d' show the same qualitative pattern as accuracy.
5. **LDI null reinterpreted**: pre-boundary lure false alarms trend up (marginal), arguing against a pure pattern-separation reading of pre-boundary memory states.
6. **SDT and FA analyses** align with hit-rate results; lure-bin gradients confirm task validity.

4.10. Robustness analyses

Outlier-trimmed RT reruns, split target/lure correctness models (ORs summarised in the conclusion), and focus on Holm-corrected boundary contrasts left the main pattern unchanged: only Item + Task Shift target post-minus-mid reached corrected significance among boundary-focused contrasts.

5. Conclusion

Phase 2 extends Report 1 with confirmatory tests on the same paired sample (recap above). Inference favoured participant-level RM ANOVA, Friedman, Holm tests, chi-square, and rank correlations, with supplementary GEE models for literature-aligned contrasts. Phase 2

achieved the stated objectives after correcting three implementation issues and adding nine planned or follow-up analyses.

Generalisation is limited by a classroom sample and moderate between-condition n ; inference spans several prespecified families (Holm within family, not across all models). Seven-item boundaries operationalise events but need not match subjective segmentation.

Main takeaways:

1. **Post-boundary cost (Item + Task Shift):** Friedman $\chi^2(2) = 11.47$, $p = .003$; post-mid $d_z = -0.477$, Holm $p = .020$; SDT post-mid $d_z = -0.473$, Holm $p = .016$.
2. **Stimulus class $\times$ boundary:** stronger for Objects in supplementary models; response direction shows general trace weakening (hits down; errors up, not only conservative bias).
3. **Carry-over null; lure FA / LDI:** familiarity-flavoured account; power rules out “too small n ” for the LDI null (contrast -0.031).
4. **Robustness:** only Item + Task Shift target post-mid survives stringent boundary-focused correction among prespecified contrasts; auxiliary GEE ORs (e.g., lure bin 1.192; Scenes 0.677–0.635) confirm MST structure dominates once covariates enter.
5. **Task quality:** lure-bin gradients intact; speed-accuracy gap stable across boundaries (≈ 0.278 s slower for incorrect).

6. Codebase and contributions

The source code for this project is available at <https://github.com/ch-pavan/brim>^{oo}.

Phase 2 was completed collaboratively:

1. **Sambu Aneesh:** Framed the confirmatory questions and statistical interpretation logic.
2. **Renu Sree Vyshnavi:** Led the results interpretation and report narrative revision.
3. **Pavan Harshit:** Implemented the slide-aligned Phase 2 pipeline, generated the simplified outputs and figures, and integrated the report assets.

References

- [1] J. M. Zacks and K. M. Swallow, “Event Segmentation,” *Current Directions in Psychological Science*, 16(2), 80-84, 2007.
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